
Multi-Fidelity Training Reshapes Uncertainty Decomposition in Atomistic Models

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Abstract

Multi-fidelity training is increasingly central to atomistic machine learning, where large low-fidelity datasets are combined with scarce high-fidelity labels. Although uncertainty estimation is critical for active learning and risk-aware deployment, how aleatoric and epistemic uncertainties behave under multi-fidelity learning remains unclear. We extend the MACE graph neural network architecture with mean-variance outputs and use deep ensembles to systematically investigate the uncertainty decomposition across single- and multi-fidelity training regimes, including system-level and energy-based distribution shifts. Our results show that high-fidelity fine-tuning does not simply improve accuracy but also changes the role of uncertainty decomposition and particularly which components are informative for generalization. Aleatoric uncertainty often loses calibration and ranking power under changes in fidelity and distribution shift, whereas epistemic uncertainty becomes a more reliable indicator of errors in unseen systems and high-energy configurations. These findings show that uncertainty estimates in multi-fidelity atomistic models are not fixed properties but change dynamically across pretraining, fine-tuning, and out-of-distribution evaluation.

1 Introduction

Uncertainty estimation plays a central role in atomistic machine learning, where models are routinely deployed beyond their training distribution, for instance in active learning, structure exploration, and molecular dynamics [1–3]. Recent work has studied uncertainty quantification in atomistic models through ensemble methods, calibration strategies, and decompositions into aleatoric and epistemic components [4–8]. In parallel, multi-fidelity learning [9–13] and transfer learning [14–17] have emerged as practical strategies for balancing computational cost and accuracy in atomistic modeling. In multi-fidelity settings, large low-fidelity (LF) datasets are combined with smaller high-fidelity (HF) ones, often corresponding to different levels of electronic-structure theory [18–20]. However, the interaction between multi-fidelity learning and uncertainty estimation remains largely unexplored.

A key missing piece is understanding how predictive uncertainty, particularly its decomposition into aleatoric and epistemic components, behaves under multi-fidelity training and distributional shifts. In standard statistical settings, this decomposition provides information on noise in the data versus uncertainty in the model [21, 22]. However, in atomistic modeling, where targets are often deterministic and data coverage is highly structured, the interpretation of these components is less clear. Moreover, existing work typically evaluates uncertainty only at convergence, overlooking how uncertainty evolves during training and how it is affected by changes in data fidelity. As a result, it remains unclear whether aleatoric and epistemic uncertainty can be used as *diagnostic signals* to understand model behavior and guide improvements in multi-fidelity settings.

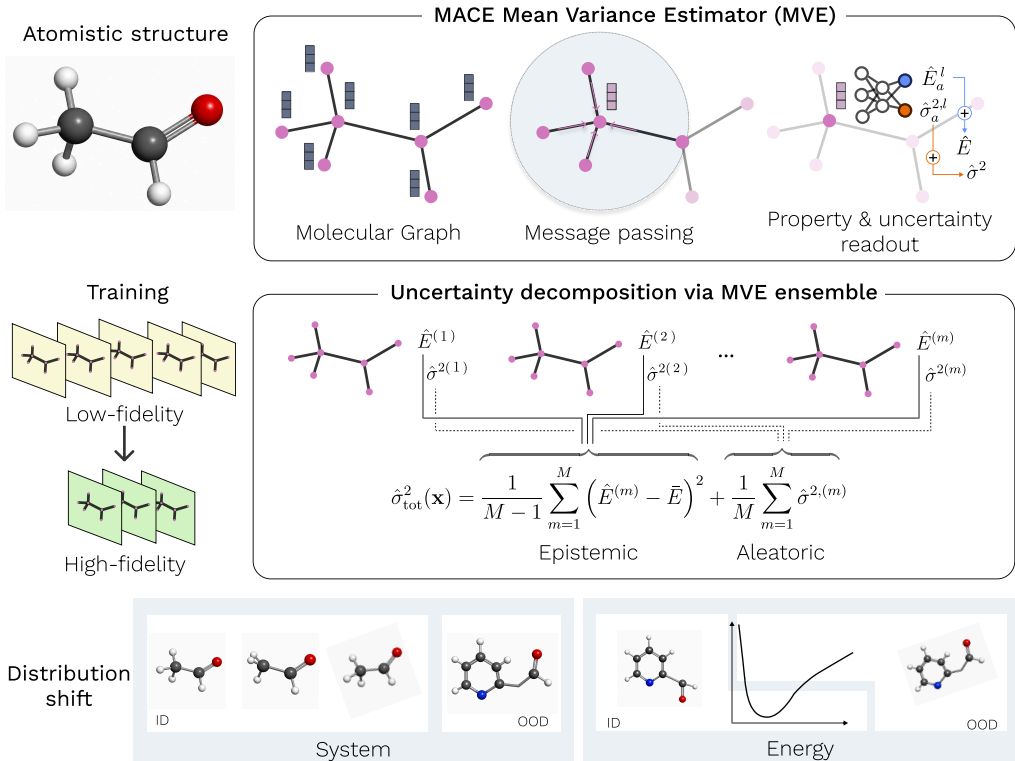


Figure 1: **Overview of the uncertainty estimation pipeline.** Atomistic structures are mapped to energy and variance predictions using a mean–variance MACE model. An ensemble of such models decomposes total predictive uncertainty into aleatoric uncertainty, given by the average predicted variance, and epistemic uncertainty, given by ensemble disagreement. We analyze these quantities across multi-fidelity training regimes and controlled system-level and energy-based distribution shifts.

Contributions. We study uncertainty in multi-fidelity atomistic models through a controlled benchmark based on ANI-1ccx dataset [23]. We analyze three fidelity regimes: low fidelity only, high fidelity, and high-fidelity finetuning, under two complementary distribution shifts: testing generalization to unseen molecular systems (system-level), and testing extrapolation to high-energy configurations within each system (energy-based). For each training protocol, we track the behavior of aleatoric and epistemic uncertainty across epochs, including their calibration and ranking capabilities. Our main contributions are: (i) we provide, to our knowledge, the first controlled analysis of aleatoric and epistemic uncertainty across multi-fidelity training, structured distribution shifts, and training dynamics; (ii) we show that uncertainty is strongly regime-dependent, with multi-fidelity fine-tuning increasing the role of epistemic uncertainty on unseen data while aleatoric uncertainty becomes poorly calibrated; (iii) we demonstrate that system-level and energy-based shifts induce qualitatively different uncertainty behavior; and (iv) we release standardized splits, protocols, and metrics to support reproducible uncertainty evaluation across fidelity regimes at the link: <https://anonymous.4open.science/r/multi-fidelity-uncertainty>.

The remainder of the paper is organized as follows. Section 2 describes the methodology, Section 3 introduces the experimental setup, Section 4 presents and discusses the results, and Section 5 concludes the paper. A detailed review of related work is provided in Appendix B.

2 Methodology

Figure 1 summarizes our uncertainty estimation pipeline. Starting from an atomistic structure $\mathbf{x}$, we use a mean–variance MACE model to predict both a total energy and an input-dependent variance. To separate uncertainty sources, we train an ensemble of independently initialized mean–variance models: the average predicted variance defines the aleatoric component, while the disagreement

between ensemble energy predictions defines the epistemic component. We then study how these uncertainty signals behave under controlled changes in training fidelity, and under two forms of distributional shift: system-level across molecular systems and energy-based across configurations of different relative energy. The following sections formalize this decomposition (Section 2.1) and describe the predictive model (Section 2.2).

2.1 Error and predictive uncertainty

For an atomistic structure (such as a molecular configuration) $\mathbf{x}$, defined by atomic positions and chemical species, and a reference property y , such as the total energy from a quantum-chemical calculation, the prediction error is the discrepancy between the model prediction $\hat{y}(\mathbf{x})$ and the reference value. This error can only be evaluated once the reference calculation is available. Predictive uncertainty, instead, aims to quantify the reliability of the prediction before observing y , and is naturally associated with the predictive distribution $p(y|\mathbf{x}, \mathcal{D})$, where $\mathcal{D}$ denotes the training dataset [24].

Predictive uncertainty is commonly decomposed into aleatoric and epistemic components [25]. Epistemic uncertainty reflects uncertainty in the predictor itself, arising from limited data coverage, model misspecification, or extrapolation beyond the training distribution. Aleatoric uncertainty is classically associated with irreducible observation noise. In this case, however, reference energies computed with a given electronic-structure method and numerical protocol are typically deterministic for a given configuration [26]. We can therefore interpret aleatoric uncertainty in a generalized, operational sense: it captures input-dependent residual variability, including heteroscedasticity and configuration-dependent difficulty.

Formally, if θ denotes the model parameters and Y the random variable associated with the reference property of structure $\mathbf{x}$, the predictive variance can be decomposed through the law of total variance [21] as

$$\text{Var}(Y|\mathbf{x}, \mathcal{D}) = \underbrace{\mathbb{E}_{p(\theta|\mathcal{D})}[\text{Var}(Y|\mathbf{x}, \theta)]}_{\text{aleatoric}} + \underbrace{\text{Var}_{p(\theta|\mathcal{D})}[\mathbb{E}(Y|\mathbf{x}, \theta)]}_{\text{epistemic}}. \quad (1)$$

The first term is the expected conditional variance of the target under a fixed model, and therefore captures variability intrinsic to the prediction task itself; for this reason, it is associated with aleatoric uncertainty. The second term measures how much the prediction changes across plausible models consistent with the training data, and therefore reflects uncertainty about the predictor; this is the epistemic component. In the following sections, we instantiate this decomposition with a mean-variance ensemble built on top of the MACE message-passing architecture.

2.2 Predictive model and uncertainty parameterization

Mean Variance MACE model. We build on MACE, an E(3)-equivariant message-passing architecture that combines message passing with higher-body atomic-cluster-expansion features for accurate and efficient modeling of the potential energy surface [27, 28]. Given an atomistic configuration $\mathbf{x}$ with atom set $\mathcal{A}$, MACE represents the total energy as a sum of atomic contributions a across interaction layers l : $\hat{E}(\mathbf{x}) = \sum_{a \in \mathcal{A}} \sum_{\ell=1}^L \hat{E}_a^\ell$.

To obtain heteroscedastic (input-dependent) uncertainty estimates, we extend the final readout layer to output, for each atom a , both an energy contribution and a variance contribution: $(\hat{E}_a^L, \hat{\sigma}_a^{2,L})$.

The structure-level predictive variance is then obtained by summing atomic variance contributions: $\hat{\sigma}^2(\mathbf{x}) = \sum_{a \in \mathcal{A}} \hat{\sigma}_a^{2,L}$.

By modeling the reference energy as a Gaussian random variable E conditioned on the structure $\mathbf{x}$: $E|\mathbf{x} \sim \mathcal{N}(\hat{E}(\mathbf{x}), \hat{\sigma}^2(\mathbf{x}))$, we train the model by minimizing the negative log-likelihood (NLL):

$$\mathcal{L}_{\text{NLL}} = \frac{1}{N} \sum_{i=1}^N \left[\frac{1}{2} \frac{(E_i - \hat{E}(\mathbf{x}_i))^2}{\hat{\sigma}^2(\mathbf{x}_i)} + \frac{1}{2} \log \hat{\sigma}^2(\mathbf{x}_i) + \frac{1}{2} \log(2\pi) \right]. \quad (2)$$

The variance learned through this objective forms the per-model estimate of aleatoric uncertainty in the ensemble decomposition below.

101 **Ensemble-based uncertainty estimation.** To capture epistemic uncertainty, we employ a deep
 102 ensemble [29] of M independently initialized mean–variance MACE models. This strategy is closely
 103 related to committee models widely used for active learning in interatomic potentials [30–32]. In our
 104 setting, each model m produces an energy and a variance: $\left(\hat{E}^{(m)}(\mathbf{x}), \hat{\sigma}^{2,(m)}(\mathbf{x})\right)$, and the ensemble
 105 predictive mean is $\bar{E}(\mathbf{x}) = \frac{1}{M} \sum_{m=1}^M \hat{E}^{(m)}(\mathbf{x})$.

106 Treating the independently trained ensemble members as approximate samples from a posterior over
 107 parameters, we approximate the two terms in the law of total variance (Equation 1) with Monte Carlo
 108 averages over the M models, yielding the following estimate of total uncertainty (TU):

$$\hat{\sigma}_{\text{tot}}^2(\mathbf{x}) = \underbrace{\frac{1}{M} \sum_{m=1}^M \hat{\sigma}^{2,(m)}(\mathbf{x})}_{\text{aleatoric}} + \underbrace{\frac{1}{M-1} \sum_{m=1}^M \left(\hat{E}^{(m)}(\mathbf{x}) - \bar{E}(\mathbf{x})\right)^2}_{\text{epistemic}}, \quad (3)$$

109 The first term (average heteroscedastic variance predicted by individual models) is interpreted as
 110 aleatoric uncertainty (AU), while the second (disagreement among independently trained ensemble
 111 members) as epistemic (EU) uncertainty [29]. This decomposition defines the three uncertainty
 112 signals analyzed throughout the paper, which we evaluate separately in terms of error ranking and
 113 calibration in order to test how their role changes across fidelity regimes and distribution shifts.

114 3 Experimental setup

115 All experiments are based on the ANI-1ccx dataset [23], which provides molecular structures
 116 evaluated at multiple levels of theory. We use DFT energies as the low-fidelity (LF) target, and
 117 coupled-cluster (CC) energies as the high-fidelity (HF) target: CC is more accurate, but substantially
 118 more expensive and therefore available for fewer configurations. These two levels of theory naturally
 119 define a multi-fidelity setting [18], where large amounts of low-cost DFT data can be combined
 120 with smaller high-accuracy CC datasets. We consider three training protocols: **LF-only**, trained
 121 on low-fidelity DFT data; **HF-only**, trained directly on CC data; and **LF→HF**, pretrained on DFT
 122 and fine-tuned on high-fidelity CC data. This protocol can also be viewed as a form of transfer
 123 learning across fidelities [18]. The **system-level shift** consists of molecular systems never seen
 124 during training, while the **energy-based shift** consists of high energy configurations of the same
 125 molecular systems seen during training. The construction of the splits, including train, validation,
 126 test-ID, test-OD is described in Appendix D (system split) and Appendix E (energy split); model
 127 and splitting hyperparameters are reported in Appendix F; computational resources are discussed in
 128 Appendix J. Our experiments are designed to answer three main research questions:

129 **RQ1: How do aleatoric and epistemic uncertainty behave at convergence across fidelity regimes?**

130 We compare models obtained from LF-only, LF→HF, and HF-only training. This allows us to test
 131 whether multi-fidelity training improves not only predictive accuracy, but also the ranking and
 132 calibration quality of uncertainty estimates.

133 **RQ2: How do uncertainty components evolve during training?** We study the evolution of
 134 aleatoric and epistemic uncertainty across epochs, including the transition from low-fidelity pretrain-
 135 ing to high-fidelity fine-tuning, to characterize how their relative contributions change across training
 136 regimes and to identify regime-dependent behaviors that are not observable at convergence.

137 **RQ3: How does the type of distribution shift affect uncertainty?** We compare system-level shift,
 138 corresponding to generalization to unseen molecular systems, with energy-based shift, corresponding
 139 to extrapolation to high-energy configurations within the same systems. This allows us to test whether
 140 structural and energetic shifts induce different AU/EU behavior.

141 3.1 Evaluation metrics

142 A central aspect of our analysis is to disentangle different properties of predictive uncertainty. In
 143 particular, we distinguish between *ranking quality*, whether uncertainty correctly identifies high-error
 144 samples, and *calibration*, whether uncertainty magnitudes match the scale of prediction errors. To this
 145 end, we employ the following complementary metrics (mathematical formulation is in Appendix C):

146 **RMSE (Root Mean Squared Error)** is the standard metric for evaluating the accuracy of atomistic
 147 models. We report RMSE normalized per atom, i.e., divided by the number of atoms in each
 148 configuration, to enable comparison across systems of different sizes.

Predicted Uncertainty is the model-predicted variance, which serves as an estimate of the expected squared error. Ideally, a well-calibrated model should produce uncertainty estimates consistent with the observed RMSE. Similar to RMSE, we report uncertainty normalized per atom to ensure consistency across configurations.

Spearman Rank Correlation measures the monotonic relationship between predicted uncertainty and prediction error. It evaluates whether uncertainty preserves the ordering of samples by difficulty, independently of its scale.

AUSE (Area Under the Sparsification Error) [33] measures how effectively uncertainty ranks samples according to their prediction error. Low AUSE indicates that uncertainty is useful for identifying unreliable predictions, with greater emphasis on correctly ranking high-error samples.

Binned Expected Normalized Calibration Error (ENCE) [34] evaluates whether predicted uncertainty magnitudes are consistent with empirical errors, and therefore measures the *calibration* of uncertainty, independently of its ranking quality.

4 Results

We now present an empirical analysis of predictive uncertainty across the experimental settings described above. Throughout this section, we analyze uncertainty along two complementary dimensions: its ability to *rank* high-error samples (measured by AUSE and Spearman correlation) and its *calibration* (measured by ENCE). This distinction is critical, as these two properties do not necessarily evolve together [35].

In Section 4.1, we analyze the final models by evaluating the behavior of aleatoric and epistemic uncertainty at convergence across fidelity regimes (RQ1). In Section 4.2, we then examine training dynamics to understand how uncertainty evolves during optimization (RQ2). In both subsections, we investigate the behavior along both system and energy distributional shifts (RQ3).

4.1 Reliability of the final models

Figure 2 reports reliability diagrams under system-level shift for three training protocols: LF-only, LF→HF, and HF-only. Each model is evaluated at the same fidelity level as its final training target: LF-only models are trained and evaluated on DFT energies, whereas LF→HF and HF-only models are evaluated on CC energies. Therefore, ID-OOD comparisons within each column isolate the effect

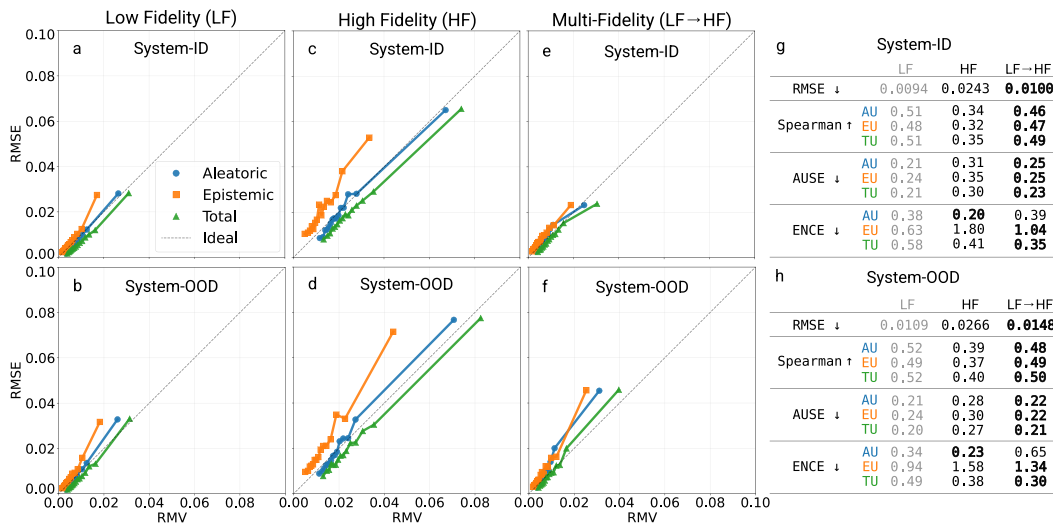


Figure 2: **Reliability diagrams under system-level shift.** Top row: in-domain test set (Test-ID). Bottom row: out-of-domain test set (Test-OOD). Columns correspond to LF-only, HF-only and LF→HF training. Gray values indicate LF-only results, which are reported on DFT targets and are therefore not directly comparable to the CC-based HF-only and LF→HF results. Bold values indicate the best-performing model among the comparable high-fidelity protocols.

of system-level distribution shift, while comparisons between LF-only and the high-fidelity protocols should be interpreted with the additional change in target fidelity in mind.

Under system-level shift, the ranking quality of the uncertainty estimates is relatively stable between Test-ID and Test-OOD. For LF-only, the total uncertainty achieves similar Spearman correlation and AUSE in both regimes, with a slight improvement in OOD. A similar trend is observed for LF→HF, where the total uncertainty improves from Test-ID to Test-OOD both in Spearman correlation and AUSE. Among the two CC-trained protocols, LF→HF clearly outperforms HF-only in predictive accuracy and ranking quality, yielding lower RMSE, higher Spearman correlation, and lower AUSE in both ID and OOD regimes. This indicates that low-fidelity pretraining provides a useful initialization for high-fidelity uncertainty estimation, whereas training directly on CC data leads to substantially weaker error ranking.

Figure 3 reports the corresponding analysis under energy-based shift, where Test-OOD consists of high-energy configurations within the same molecular systems. Compared to system-level shift, energy-based OOD is substantially more challenging: RMSE increases sharply from Test-ID to Test-OOD for all training protocols, indicating that high-energy configurations induce much larger prediction errors. Despite this increase in error, the ranking quality of the uncertainty estimates improves markedly in the OOD regime. This is visible both in Spearman correlation, which increases substantially for all protocols, and in AUSE, which decreases strongly from Test-ID to Test-OOD. For example, the total uncertainty improves from moderate ranking performance in Test-ID to much stronger error ranking in Test-OOD for LF-only, LF→HF, and HF-only models alike. This suggests that high-energy configurations produce larger and more structured errors, which are easier for uncertainty estimates to order correctly. Among the two CC-trained protocols, LF→HF again provides better predictive accuracy and better ranking quality than HF-only, supporting the role of low-fidelity pretraining in high-fidelity generalization.

However, the calibration behavior is qualitatively different from the ranking behavior. ENCE increases dramatically in the energy-OOD regime, especially for epistemic uncertainty, showing that the numerical scale of the predicted uncertainty becomes poorly calibrated even when its ordering of errors improves. Thus, energy-based shift reveals a clear separation between ranking and calibration: uncertainty estimates become better at identifying which samples are likely to have larger errors, but worse at assigning quantitatively calibrated uncertainty magnitudes. Compared with system-level shift, energy-based shift therefore induces a much stronger discrepancy between error ranking and

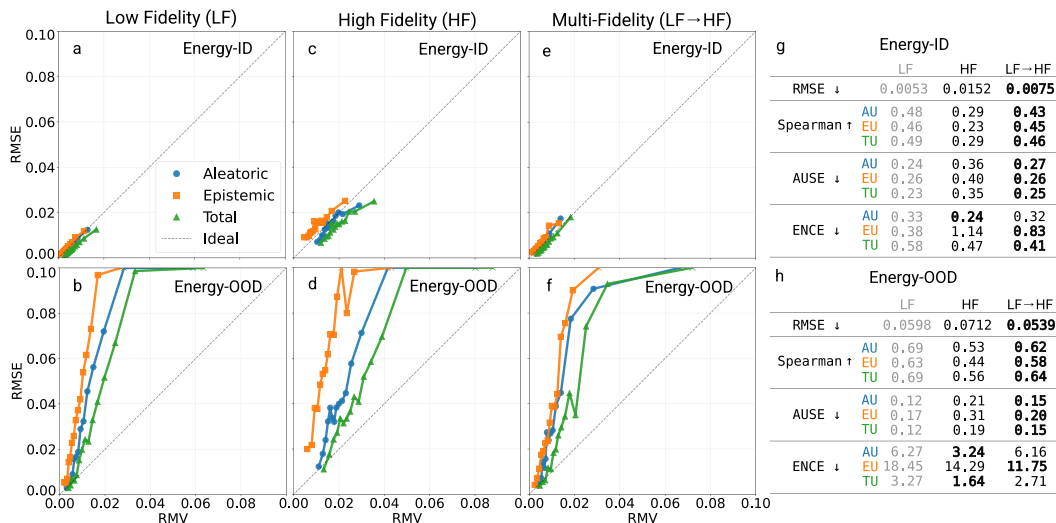


Figure 3: **Reliability diagrams under energy-based distribution shift.** Top row: in-domain test set (Test-ID). Bottom row: out-of-domain test set (Test-OOD). Columns correspond to LF-only, HF-only and LF→HF training. Gray values indicate LF-only results, which are reported on DFT targets and are therefore not directly comparable to the CC-based HF-only and LF→HF results. Bold values indicate the best-performing model among the comparable high-fidelity protocols

calibration, highlighting that the usefulness of uncertainty depends strongly on the type of distribution shift being considered.

This different behavior between system and energy distribution shift becomes even more apparent if a post-hoc recalibration is applied. In Appendix I, we additionally report reliability diagrams after isotonic calibration [35] (based on validation set). As expected, this improves the Test-ID diagrams, which become close to ideal. Since isotonic map preserves the ordering of uncertainty estimates, the ranking behavior is largely unchanged. However, its effect on scale reliability differs across OOD regimes: under system-level shift, calibrated and uncalibrated reliability remain qualitatively similar whereas under energy-based shift, a mapping learned on in-domain data leads to substantially worse uncertainty calibration for high-energy extrapolation. This provides a cautionary example: post-hoc recalibration can improve in-domain reliability while failing, or even degrading, uncertainty reliability under strong extrapolation.

4.2 Training dynamics of uncertainty

We now study how uncertainty components evolve during training. We analyze two complementary questions: first, how the roles of AU and EU change during the LF→HF transition under different distribution shifts; second, whether the high-fidelity stage of multi-fidelity training produces different uncertainty dynamics than training directly on high-fidelity data.

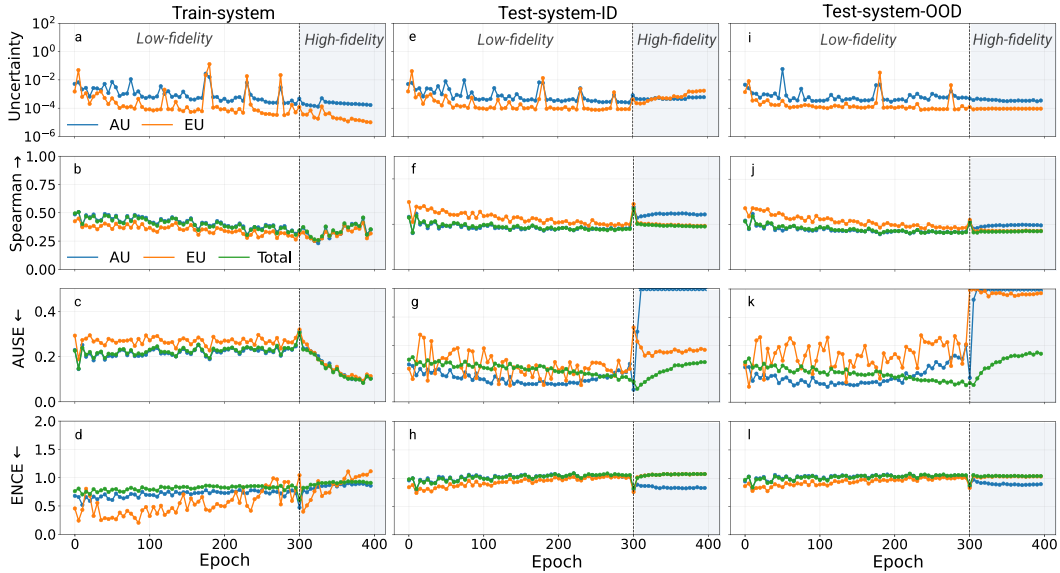


Figure 4: Training dynamics of uncertainty for LF→HF training under system-level shift. Columns correspond to evaluation on Train, Test-ID, and Test-OOD splits, respectively. Rows report uncertainty magnitude, Spearman rank correlation, AUSE, and ENCE across epochs. The dashed vertical line marks the transition from low-fidelity DFT pretraining to high-fidelity CC fine-tuning, with the shaded region indicating the high-fidelity stage. AU, EU, and total uncertainty denote aleatoric, epistemic, and summed predictive uncertainty, respectively.

Figure 4 shows the uncertainty dynamics for LF→HF training under system-level shift. During the low-fidelity pretraining phase, both AU and EU decrease in magnitude (Fig. 4a) as the model fits the DFT training objective. On the train set, AU remains the dominant uncertainty component and provides better ranking performance than EU (Fig. 4b,c), although both components worsen with training epochs. However, on unseen data (both ID and OOD), the transition to high-fidelity fine-tuning changes this behavior. AU becomes significantly less reliable: its ranking quality degrades, as shown by lower Spearman correlation (Fig. 4f,j) and higher AUSE (Fig. 4g,k), and its calibration deteriorates sharply according to ENCE (Fig. 4h,l). In contrast, after fine-tuning, EU becomes more informative than AU on unseen data (Fig. 4f,g,h), while total uncertainty provides the most stable signal across regimes and maintains accurate ranking even under OOD evaluation (AUSE in Fig. 4j,k). This suggests that the transition from LF pretraining to HF fine-tuning changes the relative role

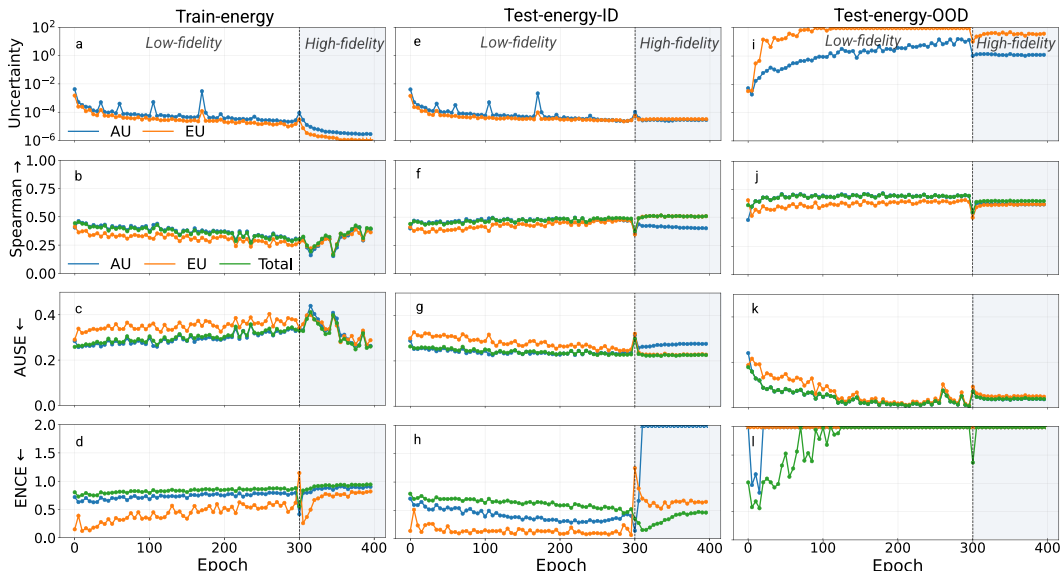


Figure 5: Training dynamics of uncertainty for LF→HF training under energy-based shift. Columns correspond to evaluation on Train, Test-ID, and Test-OOD splits, respectively. Rows report uncertainty magnitude, Spearman rank correlation, AUSE, and ENCE across epochs. The dashed vertical line marks the transition from low-fidelity DFT pretraining to high-fidelity CC fine-tuning, with the shaded region indicating the high-fidelity stage. AU, EU, and total uncertainty denote aleatoric, epistemic, and summed predictive uncertainty, respectively.

of AU and EU: AU remains effective on fitted training data, whereas EU becomes more relevant for generalization. Overall, system-level shift shows that high-fidelity fine-tuning redistributes uncertainty informativeness from the aleatoric component toward the epistemic component, while total uncertainty remains the most robust diagnostic signal.

In Figure 5 we analyze the training dynamics of uncertainty for the training/test splits associated with the energy-based shift. On Train and Test-ID, AU and EU follow a behavior similar to the system-level setting, with their magnitudes decreasing during low-fidelity pretraining (Fig. 5a,e). Similarly, high-fidelity fine-tuning makes AU less reliable on unseen data with similar energies (Test-ID) across all metrics (Fig. 5f,g,h). Instead, when considering higher-energy configurations (Test-OOD) we observe a qualitatively different uncertainty dynamics. Indeed, the absolute value of uncertainty increases by orders of magnitude during pretraining, with EU becoming the dominant component (Fig. 5i). This indicates that, as the model specializes to the low-energy training manifold, high-energy configurations are increasingly recognized as epistemically uncertain (Fig. 5i). At the same time, ranking quality improves substantially in the OOD regime relative to the ID regime, as shown by higher Spearman correlation (Fig. 5j vs f) and lower AUSE (Fig. 5k vs g). Thus, uncertainty becomes better at ordering high-error samples under energy-OOD. However, ENCE remains very large (Fig. 5l), indicating that improved ranking does not imply calibrated uncertainties.

Taken together, the system-level and energy-based shifts reveal a common pattern: after fine-tuning on high-fidelity data, the uncertainty associated with residual variability becomes less reliable on unseen data, while ensemble disagreement becomes the more informative signal for generalization. In both cases, the total uncertainty is the most robust and reliable measure, since it combines both effects and remains useful for ranking errors across evaluation regimes. The two distribution shifts differ mainly in how clearly they expose this behavior. System-level shift tests transfer to new molecular systems, where errors increase only moderately and the uncertainty signals change more subtly. Energy-based shift is a stronger extrapolation test: high-energy configurations are not necessarily noisier targets, but poorly supported regions of configuration space. In this regime, ensemble disagreement increases sharply and more directly identifies where the model is extrapolating. Thus, energy-based shift provides a clearer example of epistemic uncertainty becoming the relevant signal, whereas aleatoric uncertainty remains tied to the fitted in-domain regime.

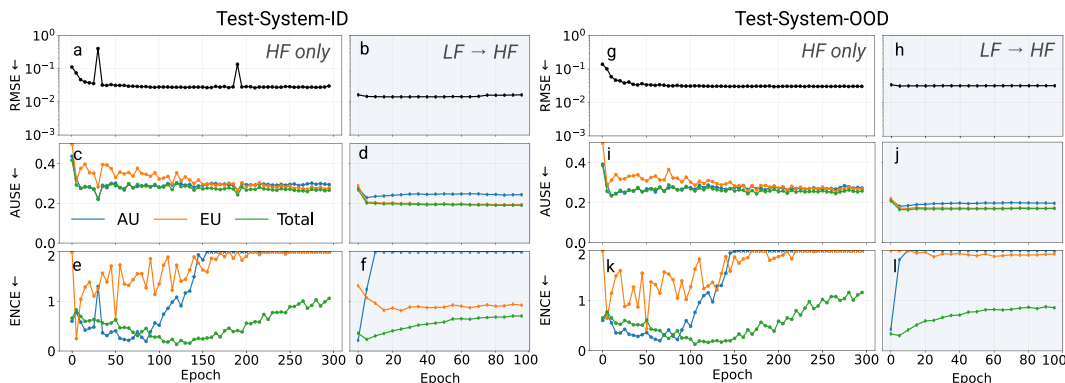


Figure 6: **Training dynamics of uncertainty for HF-only vs high-fidelity fine-tuning.** Columns compare Test-ID and Test-OOD evaluation regimes under system-level shift; within each regime, the left block shows HF-only training from scratch and the right block shows the high-fidelity fine-tuning stage initialized from the LF pretrained model.

Finally, Figure 6 directly compares HF-only training with the high-fidelity fine-tuning stage of LF→HF training under system-level shift both in terms of accuracy (RMSE) and uncertainty ranking (AUSE) and calibration (ENCE), showing how they vary across epochs for Test-ID and Test-OOD. As expected, LF→HF yields slightly better predictive accuracy than HF-only and reaches lower RMSE more efficiently in both evaluation regimes (Fig. 6a vs b, g vs h). However, the larger difference appears in uncertainty quality: LF→HF achieves better error ranking and calibration, with lower AUSE (Fig. 6c vs d, i vs j) and ENCE (Fig. 6e vs f, k vs l), and shows substantially more stable trajectories during training. Thus, the benefit of multi-fidelity learning is not limited to accuracy. Low-fidelity pretraining appears to regularize the subsequent high-fidelity adaptation, producing uncertainty estimates that are both more reliable and stable.

Additional results are provided in the Appendix. In Appendix G, we present a more detailed analysis of energy-based shift by partitioning configurations into per-system energy quantiles. Appendix H reports a comprehensive comparison of RMSE, loss, and uncertainty quality metrics across all experimental configurations. Finally, Appendix K discusses limitations.

5 Conclusion

We examined aleatoric (AU) and epistemic (EU) uncertainty in deep ensembles across multi-fidelity atomistic models, varying data fidelity, distribution shift, and training trajectory. Across these settings, our results show that uncertainty decomposition is not fixed: the relative usefulness of AU and EU changes with the training pathway and with the type of distribution shift. At convergence (**RQ1**), low-fidelity pretraining followed by high-fidelity fine-tuning consistently yielded better predictive accuracy, error ranking, and calibration than training directly on high-fidelity data, indicating that multi-fidelity learning also acts as a regularizer for uncertainty. Epistemic uncertainty emerged as the more informative signal on unseen data, while aleatoric uncertainty remained reliable primarily on the fitted in-domain regime. Training dynamics (**RQ2**) revealed that this shift is triggered by the transition to high-fidelity fine-tuning: on the training set AU remained dominant and well-calibrated, but on unseen data it degraded, whereas EU became the stronger error-ranking component. The total (aleatoric + epistemic) uncertainty, however, provided the most robust signal across all regimes and throughout training. The type of distribution shift (**RQ3**) strongly affected uncertainty behavior. System-level shift, corresponding to unseen molecular systems, produced moderate changes in uncertainty behavior. Energy-based shift, corresponding to high-energy configurations within known systems, exposed a stronger extrapolative regime: epistemic uncertainty increased sharply and substantially improved error ranking, while calibration deteriorated. Thus, under strong extrapolation, even if uncertainty magnitudes may be poorly calibrated, uncertainty remains useful for ranking difficult configurations. Overall, our findings show that uncertainty decomposition in atomistic models is not static, but depends on the training pathway, the fidelity transition, and the distribution shift under evaluation.

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A Liquid water

- afqmc: 200 configs -> train=130, val=20, test=50; no nonzero forces
- blyp: 531 configs -> train=345, val=53, test=133; 531 configs with nonzero forces
- ccscd: 200 configs -> train=130, val=20, test=50; no nonzero forces
- ccscdt: 200 configs -> train=130, val=20, test=50; no nonzero forces
- hf: 531 configs -> train=345, val=53, test=133; 531 configs with nonzero forces
- revpbe-d3: 531 configs -> train=345, val=53, test=133; 531 configs with nonzero forces

A.1 Reliability

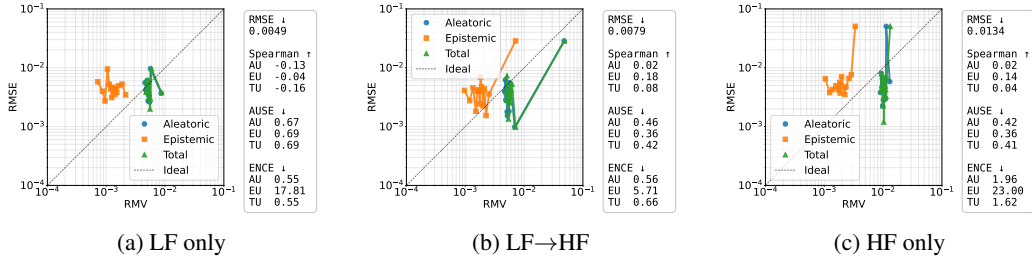
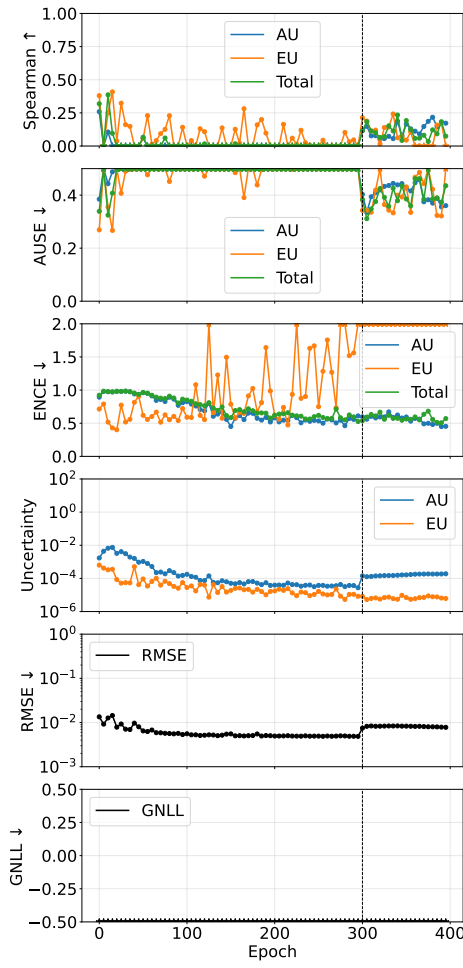
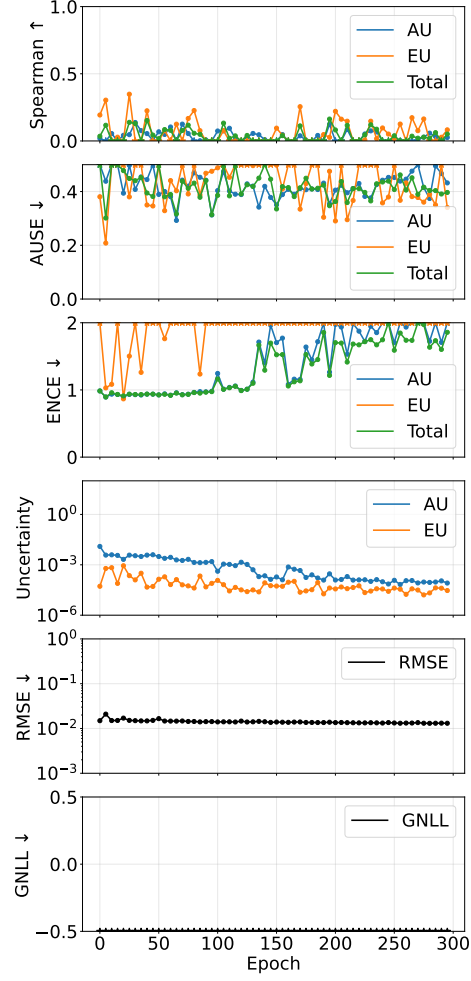


Figure 7: Reliability diagram test set.

A.2 Train dynamics



(a) LF→HF



(b) HF only

B Related Work

Recent surveys and perspectives provide broad overviews of the Uncertainty quantification (UQ) field, covering Bayesian methods, ensembles, calibration strategies, active learning, and the relationship between uncertainty, dataset composition, transferability, and model robustness [4, 5, 36]. These works highlight that benchmarking and interpretation of UQ methods remain open challenges, especially in transfer and multi-fidelity settings.

A large body of work has focused on practical uncertainty estimators for neural network potentials. Ensemble-based methods remain among the most widely used and robust approaches. For instance, Tan et al. [6] compare ensembles against single-model uncertainty methods such as mean-variance estimation, evidential regression, and Gaussian mixture models, finding that no method uniformly dominates across datasets and metrics, while ensembles remain strong for robustness and generalization. Other work has proposed more efficient ensemble variants, including multi-headed architectures, committee models, shallow or last-layer ensembles, and shared-representation ensembles, to reduce computational cost while retaining useful uncertainty estimates [37–39].

Several works have specifically addressed calibration and uncertainty decomposition in atomistic models. Busk et al. [8] propose a framework for calibrated aleatoric and epistemic uncertainty in graph neural network potentials, using deep ensembles to decompose total uncertainty and post-hoc recalibration to improve agreement between predicted uncertainty and observed error. More generally, Valdenegro-Toro and Mori [7] study disentangled aleatoric and epistemic uncertainty across different UQ methods, showing that the interaction between these components can be non-trivial, that

aleatoric uncertainty may become unreliable under distribution shift, and that ensembles provide the most consistent decomposition. In molecular property prediction, Heid et al. [40] further analyze uncertainty sources by separating epistemic uncertainty into bias and variance contributions, showing that different sources of error dominate depending on noise level, dataset size, model architecture, and representation.

A related line of work studies multi-fidelity learning in atomistic modeling. Gardner et al. [12] investigate multi-fidelity training strategies for atomistic models, including pretraining–fine-tuning and multi-headed training, showing that fine-tuning can improve accuracy but that learned representations may remain method-specific, while multi-headed models can learn more method-agnostic representations across fidelities. More broadly, multi-fidelity UQ has been studied in surrogate modeling and materials design, often using Gaussian process models to combine information across fidelity levels while quantifying predictive confidence [10].

Finally, the interpretation of uncertainty depends strongly on the evaluation metric. Rasmussen et al. [41], Lind et al. [33] analyze common regression UQ metrics, including AUSE, calibration error, Spearman correlation, and NLL, showing that they capture different properties and can lead to different conclusions.

Overall, existing work has made substantial progress on uncertainty estimation for atomistic models, uncertainty decomposition, calibration, efficient ensemble methods, and multi-fidelity training. However, these directions have largely been studied separately. In contrast, our work provides a controlled empirical analysis of aleatoric and epistemic uncertainty across multi-fidelity training regimes, system-level and energy-based distribution shifts, and training dynamics. This allows us to study not only whether uncertainty is accurate at convergence, but also how its components evolve and whether they can serve as diagnostic signals for model behavior under changing fidelity and data regimes.

C Details on uncertainty evaluation metrics

In this section, we provide the formal definitions of the metrics used in the main text. These metrics are designed to capture complementary aspects of predictive uncertainty, including ranking quality, calibration, and probabilistic correctness.

C.1 Area under the sparsification error (AUSE)

AUSE evaluates how well uncertainty ranks samples according to their prediction errors.

Let $S = \{(x_i, y_i)\}_{i=1}^N$ be a test set, and define the mean absolute error:

$$\text{MAE}(S) = \frac{1}{|S|} \sum_{(x_i, y_i) \in S} |y_i - \hat{y}(x_i)|. \quad (4)$$

For a fraction $\alpha \in [0, 1]$, let:

- $S_V(\alpha)$ denotes the subset obtained by removing the top- α fraction of samples with the largest prediction errors (oracle ranking),
- $S_U(\alpha)$ denote the subset obtained by removing the top- α fraction of samples with the largest predicted uncertainties.

The sparsification error compares the degradation in performance when removing samples based on uncertainty versus the oracle ranking. AUSE is defined as:

$$\text{AUSE}(S) = \frac{1}{\text{MAE}(S)} \int_0^1 (\text{MAE}(S_U(\alpha)) - \text{MAE}(S_V(\alpha))) d\alpha. \quad (5)$$

AUSE measures the *ranking quality* of uncertainty. Low AUSE indicates that uncertainty successfully identifies high-error samples. Compared to pure ordering metrics such as Spearman correlation, AUSE is sensitive to the *magnitude* of errors, penalizing misranking of large-error samples more strongly.

488 **C.2 Spearman rank correlation**

489 Spearman’s rank correlation evaluates the monotonic relationship between uncertainty and prediction
 490 error:

$$\rho_S = \text{Corr}_{\text{rank}}(\sigma^2(x), e^2(x)), \quad (6)$$

491 where $e(x) = y - \hat{y}(x)$.

492 This metric measures whether uncertainty preserves the ordering of errors. It is invariant to mono-
 493 tonic transformations of the uncertainty scale (e.g., post-hoc calibration), and therefore isolates the
 494 *informativeness* of uncertainty independently of its magnitude.

495 **C.3 Binned expected normalized calibration error (ENCE)**

496 Calibration evaluates whether predicted uncertainty magnitudes match empirical errors.

497 We partition the test set into B bins based on predicted uncertainty. For each bin b , we compute:

$$\text{RMSE}_b = \sqrt{\mathbb{E}[e^2|b]}, \quad \text{RMV}_b = \sqrt{\mathbb{E}[\sigma^2|b]}. \quad (7)$$

498 The Expected Normalized Calibration Error (ENCE) is defined as:

$$\text{ENCE} = \frac{1}{B} \sum_{b=1}^B \frac{|\text{RMV}_b^2 - \text{RMSE}_b^2|}{\text{RMV}_b^2}. \quad (8)$$

499 ENCE measures the *scale correctness* of uncertainty. Unlike AUSE and Spearman, it does not assess
 500 ranking quality, but instead quantifies whether predicted variances match observed squared errors.

501 As a widely adopted standard, we set B equal to 15 in this work.

502 **C.4 Negative log-likelihood (NLL)**

503 NLL evaluates the probabilistic correctness of the predictive distribution:

$$\text{NLL}(S) = -\frac{1}{N} \sum_{i=1}^N \log p(y_i|x_i). \quad (9)$$

504 Under the Gaussian predictive assumption:

$$\text{NLL} = \frac{1}{2} \left(\frac{e^2}{\sigma^2} + \log \sigma^2 \right) + C, \quad (10)$$

505 where C is a constant.

506 NLL is a strictly proper scoring rule and jointly evaluates calibration and sharpness. However, it does
 507 not isolate ranking behavior and can be dominated by a small number of poorly calibrated predictions.

508 **D System-level data splitting protocol**

509 **Dataset structure.** The ANI-1x dataset consists of approximately 3,000 molecular systems, each
 510 associated with a variable number of molecular configurations (geometries). For each configuration,
 511 multiple quantum-chemical annotations may be available, including density functional theory (DFT)
 512 energies and, for a subset, high-fidelity coupled-cluster energies (CCSD(T)/CBS).

513 D.1 System-OOD split

514 Let $\mathcal{S}$ denote the set of molecular systems. We first perform a random partition of $\mathcal{S}$ into two disjoint
515 subsets:

$$\mathcal{S} = \mathcal{S}_{\text{seen}} \cup \mathcal{S}_{\text{OOD}}, \quad \mathcal{S}_{\text{seen}} \cap \mathcal{S}_{\text{OOD}} = \emptyset, \quad (11)$$

516 where $\mathcal{S}_{\text{OOD}}$ contains a fixed proportion (e.g., 30%) of systems and is reserved exclusively for
517 *System-OOD testing*. All configurations belonging to systems in $\mathcal{S}_{\text{OOD}}$ are excluded from training
518 and validation.

519 D.2 In-domain (ID) split within seen systems

520 The remaining systems $\mathcal{S}_{\text{seen}}$ define the in-domain (ID) regime. For each system $s \in \mathcal{S}_{\text{seen}}$, we
521 randomly shuffle its configurations and partition them into three disjoint subsets according to fixed
522 proportions:

$$\mathcal{X}_s = \mathcal{X}_s^{\text{train}} \cup \mathcal{X}_s^{\text{val}} \cup \mathcal{X}_s^{\text{test-ID}}, \quad (12)$$

523 where $\mathcal{X}_s^{\text{train}}$, $\mathcal{X}_s^{\text{val}}$, and $\mathcal{X}_s^{\text{test-ID}}$ denote the training, validation, and in-domain test configurations of
524 system s , respectively.

525 Aggregating across systems yields:

$$\mathcal{X}_{\text{ID}}^{\text{train}} = \bigcup_{s \in \mathcal{S}_{\text{seen}}} \mathcal{X}_s^{\text{train}}, \quad (13)$$

$$\mathcal{X}_{\text{ID}}^{\text{val}} = \bigcup_{s \in \mathcal{S}_{\text{seen}}} \mathcal{X}_s^{\text{val}}, \quad (14)$$

$$\mathcal{X}_{\text{ID}}^{\text{test}} = \bigcup_{s \in \mathcal{S}_{\text{seen}}} \mathcal{X}_s^{\text{test-ID}}. \quad (15)$$

526 This procedure ensures that: (i) no configuration leakage occurs across splits, and (ii) the in-domain
527 test set contains unseen configurations from the same distribution of molecular systems as training.

528 D.3 Annotation-specific dataset construction

529 From each configuration split, we construct task-specific datasets:

- 530 • **DFT datasets:** configurations with available DFT annotations are selected from each split,
531 optionally using stratified sampling to prevent over-representation of large systems.
- 532 • **CC datasets:** high-fidelity CC configurations are selected *only from the already selected*
533 *DFT configurations*, ensuring

$$\mathcal{X}_{\text{CC}} \subseteq \mathcal{X}_{\text{DFT}}, \quad (16)$$

534 which guarantees consistency for multi-fidelity pretraining and fine-tuning.

535 D.4 Stratified sampling across molecular systems

536 After defining the system-level splits, we optionally subsample configurations within each split to
537 reach fixed target sizes (e.g., $n_{\text{DFT}}^{\text{train}}$, $n_{\text{DFT}}^{\text{val}}$). To avoid over-representation of large systems (some of
538 which contain orders of magnitude more configurations than others) we employ a *stratified sampling*
539 *scheme at the system level*.

540 **Available pools.** For a given split $\sigma \in \{\text{train, val, test-ID, test-OOD}\}$ and theory level (DFT or CC),
541 let

$$\mathcal{X}_{s,\sigma} \quad (17)$$

542 denote the set of configurations of system s that: (i) belong to split σ , and (ii) have a valid annotation
543 for the selected theory level. Let

$$n_{s,\sigma} = |\mathcal{X}_{s,\sigma}| \quad (18)$$

544 be the number of available candidates for system s in split σ . The total number of available
545 configurations in that split is

$$N_{\sigma} = \sum_s n_{s,\sigma}. \quad (19)$$

546 **Quota allocation.** Given a desired target size $N_\sigma^{\text{target}} \leq N_\sigma$, we allocate per-system quotas $q_{s,\sigma}$
 547 proportionally to the number of available configurations:

$$\tilde{q}_{s,\sigma} = N_\sigma^{\text{target}} \frac{w_{s,\sigma}}{\sum_{s'} w_{s',\sigma}}, \quad (20)$$

548 where the weights are defined as

$$w_{s,\sigma} = \begin{cases} n_{s,\sigma}, & \text{if no cap is applied,} \\ \min(n_{s,\sigma}, \kappa), & \text{with optional cap } \kappa. \end{cases} \quad (21)$$

549 The optional cap κ limits the influence of very large systems while still allowing them to contribute
 550 more samples when necessary.

551 The final integer quotas $q_{s,\sigma}$ are obtained by:

- 552 1. taking the floor of $\tilde{q}_{s,\sigma}$, and
 - 553 2. distributing the remaining samples to systems with the largest fractional remainders,
- 554 ensuring that

$$\sum_s q_{s,\sigma} = N_\sigma^{\text{target}}. \quad (22)$$

555 **Sampling without replacement.** For each system s , we then draw $q_{s,\sigma}$ configurations uniformly at
 556 random without replacement from $\mathcal{X}_{s,\sigma}$. This yields the final stratified subset

$$\hat{\mathcal{X}}_\sigma = \bigcup_s \hat{\mathcal{X}}_{s,\sigma}, \quad (23)$$

557 with $|\hat{\mathcal{X}}_\sigma| = N_\sigma^{\text{target}}$.

558 **Multi-fidelity constraint.** For the high-fidelity CC dataset, candidate configurations are first
 559 restricted to those already selected in the corresponding DFT split, enforcing

$$\mathcal{X}_\sigma^{\text{CC}} \subseteq \hat{\mathcal{X}}_\sigma^{\text{DFT}}. \quad (24)$$

560 Stratified quota allocation is then applied within this restricted pool, guaranteeing consistency between
 561 low- and high-fidelity datasets.

562 **Properties.** This procedure:

- 563 • preserves the system-level split,
- 564 • prevents dominance of large systems,
- 565 • enforces exact target sizes,
- 566 • maintains $\text{CC} \subseteq \text{DFT}$,
- 567 • and ensures reproducibility via a fixed random seed.

568 D.5 Resulting evaluation regimes

569 This protocol produces four disjoint evaluation settings:

- 570 • **Train (ID):** configurations from $\mathcal{S}_{\text{seen}}$ used for model optimization.
- 571 • **Validation (ID):** configurations from $\mathcal{S}_{\text{seen}}$ used for early stopping and uncertainty calibration.
- 572 • **Test-ID:** unseen configurations from $\mathcal{S}_{\text{seen}}$.
- 573 • **Test-OOD (System-OOD):** configurations from $\mathcal{S}_{\text{OOD}}$, representing novel molecular systems.
- 574 •
- 575 •

576 By construction, the System-OOD evaluation measures generalization to entirely unseen molecular
 577 systems, while the Test-ID split quantifies generalization within the same system distribution with-
 578 out configuration leakage. This split design enables a controlled comparison between in-domain
 579 performance and true structural distributional shift.

E Energy-Based data splitting protocol

To evaluate generalization across energetic regimes within each molecular system, we construct an energy-based split that separates low-energy (in-domain) configurations from high-energy (out-of-domain) configurations.

Per-system relative energies. For each molecular system s , let $\mathcal{X}_s$ denote its full set of configurations. Given an energy annotation $E_s(x)$ (by default the DFT energy), we compute relative energies

$$\Delta E_s(x) = E_s(x) - \min_{x' \in \mathcal{X}_s} E_s(x'). \quad (25)$$

Relative energies are computed independently for each system to remove absolute offsets.

Empirical energy quantiles. Let $\{x_i\}_{i=1}^{N_s}$ be the configurations of system s with valid energies. We assign to each configuration an empirical quantile rank

$$q_s(x_i) = \frac{\text{rank}(\Delta E_s(x_i))}{N_s - 1}, \quad (26)$$

where ranks are computed after sorting ΔE_s in ascending order. This yields $q_s(x) \in [0, 1]$ for all valid configurations.

Energy domains. Given two thresholds $q_{\text{low}} < q_{\text{high}}$, we define:

$$\mathcal{X}_s^{\text{low}} = \{x \in \mathcal{X}_s : q_s(x) \leq q_{\text{low}}\}, \quad (27)$$

$$\mathcal{X}_s^{\text{high}} = \{x \in \mathcal{X}_s : q_s(x) \geq q_{\text{high}}\}. \quad (28)$$

Configurations with $q_{\text{low}} < q_s(x) < q_{\text{high}}$ form an intermediate band that is excluded from all splits to ensure a clear separation between in-domain and out-of-domain samples.

Low-energy ID splits. For each system, the low-energy set $\mathcal{X}_s^{\text{low}}$ is randomly shuffled and partitioned into:

$$\mathcal{X}_s^{\text{low}} = \mathcal{X}_s^{\text{train}} \cup \mathcal{X}_s^{\text{val}} \cup \mathcal{X}_s^{\text{test-ID}}, \quad (29)$$

according to fixed proportions $(p_{\text{train}}, p_{\text{val}}, p_{\text{test-ID}})$. Aggregating across systems yields:

$$\mathcal{X}_{\text{ID}}^{\text{train}} = \bigcup_s \mathcal{X}_s^{\text{train}}, \quad (30)$$

$$\mathcal{X}_{\text{ID}}^{\text{val}} = \bigcup_s \mathcal{X}_s^{\text{val}}, \quad (31)$$

$$\mathcal{X}_{\text{ID}}^{\text{test-ID}} = \bigcup_s \mathcal{X}_s^{\text{test-ID}}. \quad (32)$$

Energy-OOD set. All high-energy configurations are assigned to the OOD pool:

$$\mathcal{X}^{\text{test-OOD}} = \bigcup_s \mathcal{X}_s^{\text{high}}. \quad (33)$$

Theory-specific datasets. For each split $\sigma \in \{\text{train}, \text{val}, \text{test-ID}, \text{test-OOD}\}$:

- DFT configurations are selected from $\mathcal{X}_\sigma$ using stratified sampling across systems.
- CC configurations are selected exclusively from the corresponding DFT subset, enforcing

$$\mathcal{X}_\sigma^{\text{CC}} \subseteq \mathcal{X}_\sigma^{\text{DFT}}. \quad (34)$$

Properties. This protocol:

- defines energy shift independently within each molecular system,
- avoids configuration leakage across splits,
- enforces a strict low/high separation,
- preserves per-system diversity via stratified sampling,
- guarantees $\text{CC} \subseteq \text{DFT}$,
- and is fully reproducible under a fixed random seed.

609 **F Experimental hyperparameters**

610 **F.1 Dataset construction parameters**

611 The ANI-1x dataset was processed with the following settings.

612 **Theory levels.**

- 613 • Low-fidelity (DFT): `wb97x_tz.energy`.
- 614 • High-fidelity (CC): `ccsd(t)_cbs.energy`.
- 615 • Forces were not used for either fidelity level.

616 **System-level split.**

- 617 • Fraction of systems seen during training: $p_{\text{seen}} = 0.6$.
- 618 • Fraction of out-of-distribution (OOD) systems: 0.4.

619 **Configuration split within seen systems.**

- 620 • Train: $p_{\text{train}} = 0.6$.
- 621 • Validation: $p_{\text{val}} = 0.2$.
- 622 • Test-ID (in-distribution): $p_{\text{test-ID}} = 0.2$.

623 **Stratified sampling.**

- 624 • Maximum per-system soft cap: $\kappa = 64$.
- 625 • Random seed: 0.

626 **Target dataset sizes.** The DFT and CC datasets were drawn to the sizes in Table 1. By construction,
627 each CC split is a strict subset of the corresponding DFT split.

Table 1: Number of configurations per split for DFT and CC levels.

Split	DFT size	CC size
Train	50,000	5,000
Validation	10,000	1,000
Test-ID	50,000	5,000
Test-OOD	50,000	5,000

628 **F.2 Model architecture and training configuration**

629 All experiments used the MACE architecture with the following setup.

630 **Model architecture.**

- 631 • Model: MACE.
- 632 • Number of channels: 128.
- 633 • Maximum spherical harmonic order: $L_{\text{max}} = 1$.
- 634 • Cutoff radius: $r_{\text{max}} = 5.0 \text{ \AA}$.
- 635 • Atomic energy baseline (E_0): average over the training set.
- 636 • Ensemble size M : 10.

637 **Training setup.**

- 638 • Loss function: Gaussian negative log-likelihood.
- 639 • Predictive mode: energy only (forces not used).
- 640 • Forces weight: 0.0.
- 641 • Batch size: 64.
- 642 • Maximum epochs: 300.
- 643 • Stage-two loss weight (SWA): disabled.

644 G A deeper look into energy shift

645 Figure 9 provides a finer-grained view of the energy-based OOD regime by stratifying high-energy
 646 configurations into within-system energy quantiles. Across all training protocols, uncertainty in-
 647 creases with relative energy, confirming that high-energy configurations correspond to progressively
 648 more difficult and less supported regions of configuration space. This trend is especially pronounced
 649 in the highest-energy bin, where uncertainty increases by several orders of magnitude.

650 The AU/EU balance depends strongly on the training protocol. For LF-only models, EU dominates
 651 in the highest-energy bin, indicating that extrapolation to extreme energetic regimes is primarily
 652 captured as model uncertainty. LF→HF shows a similar trend, with both AU and EU increasing
 653 sharply at high energy and EU becoming dominant in the most extreme bin. In contrast, HF-only
 654 exhibits a weaker increase across intermediate quantiles, but again shows a sharp rise of EU in the
 655 highest-energy bin.

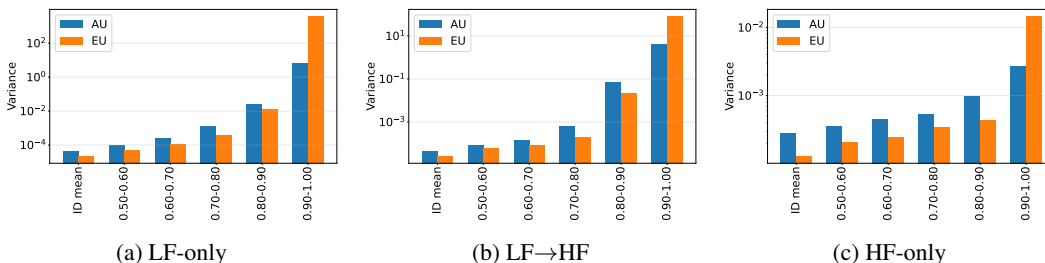


Figure 9: Mean aleatoric uncertainty (AU) and epistemic uncertainty (EU) as a function of within-system relative energy in the energy-OOD setting. For each subfigure, the leftmost group reports the system-balanced mean uncertainty on the ID test split (first half of the quantile). The remaining groups partition the OOD test split into energy-quantile bins defined separately within each molecular system, so that energies are compared only relative to configurations of the same system. For each bin, uncertainties are first averaged over configurations within each system and then averaged across systems, preventing systems with more configurations from dominating the estimate. The y-axis is shown on a logarithmic scale.

656 H Comparison of training regimes at fixed distributional shift

657 Figures 10–15 provide an alternative view of the results by comparing training regimes (LF-only,
 658 LF→HF, and HF-only) at fixed combinations of dataset (Train-ID, Test-ID, and Test-OOD) and
 659 distributional shift (system-level and energy-based). Each figure reports Spearman correlation, AUSE,
 660 ENCE, uncertainty magnitude, RMSE, and Gaussian negative log-likelihood (GNLL).

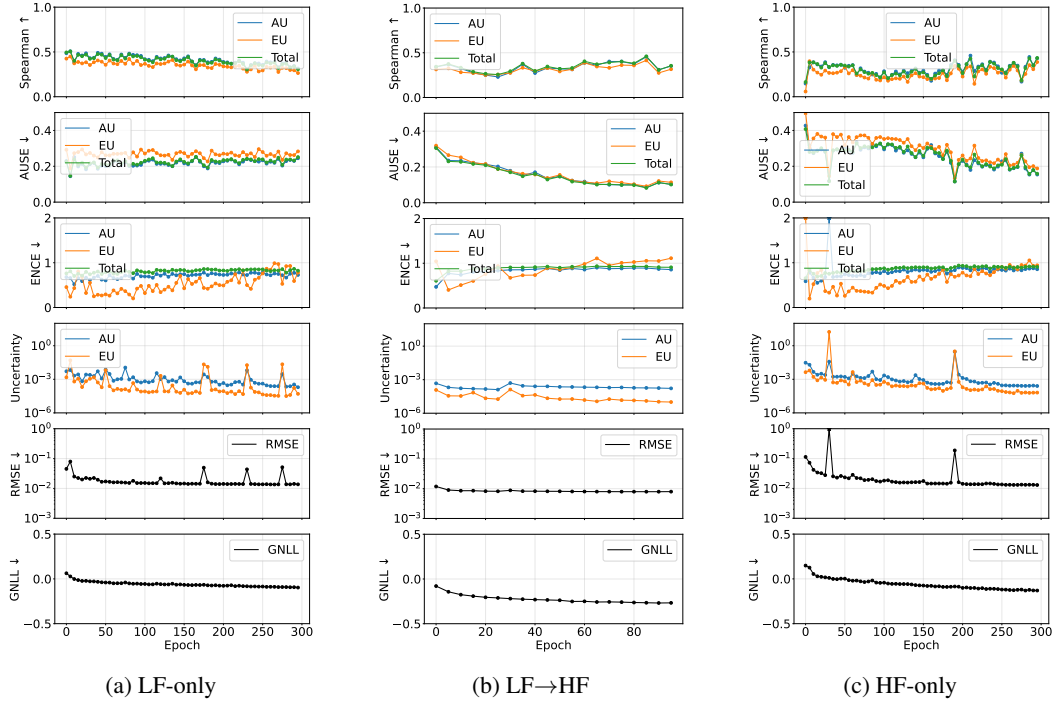


Figure 10: Training dynamics of uncertainty under system-level shift, evaluated on the training set (Train-ID). Columns compare LF-only, LF→HF, and HF-only training.

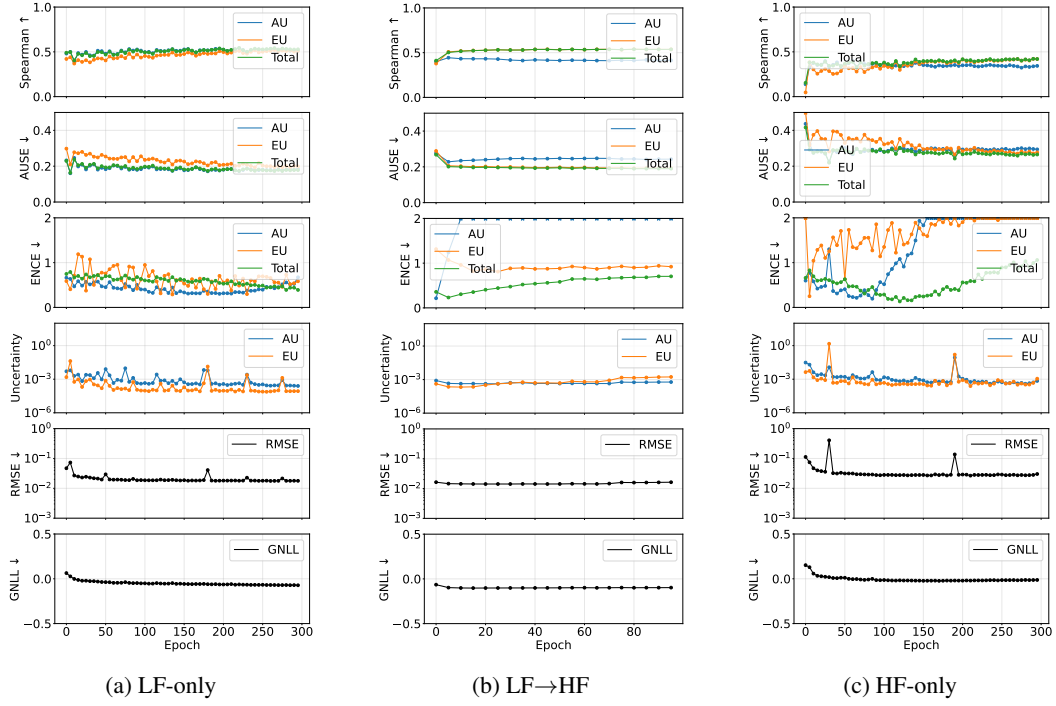


Figure 11: Training dynamics of uncertainty under system-level shift, evaluated on the in-domain test set (Test-ID). Columns compare LF-only, LF→HF, and HF-only training.

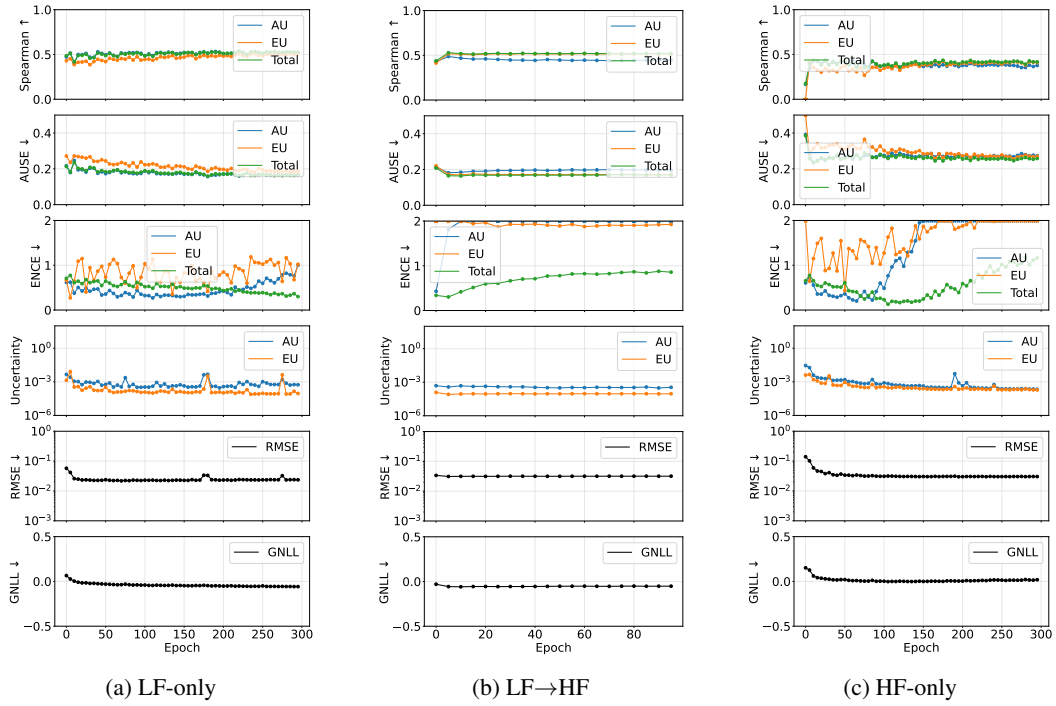


Figure 12: Training dynamics of uncertainty under system-level shift, evaluated on the out-of-domain test set (Test-OOD). Columns compare LF-only, LF→HF, and HF-only training.

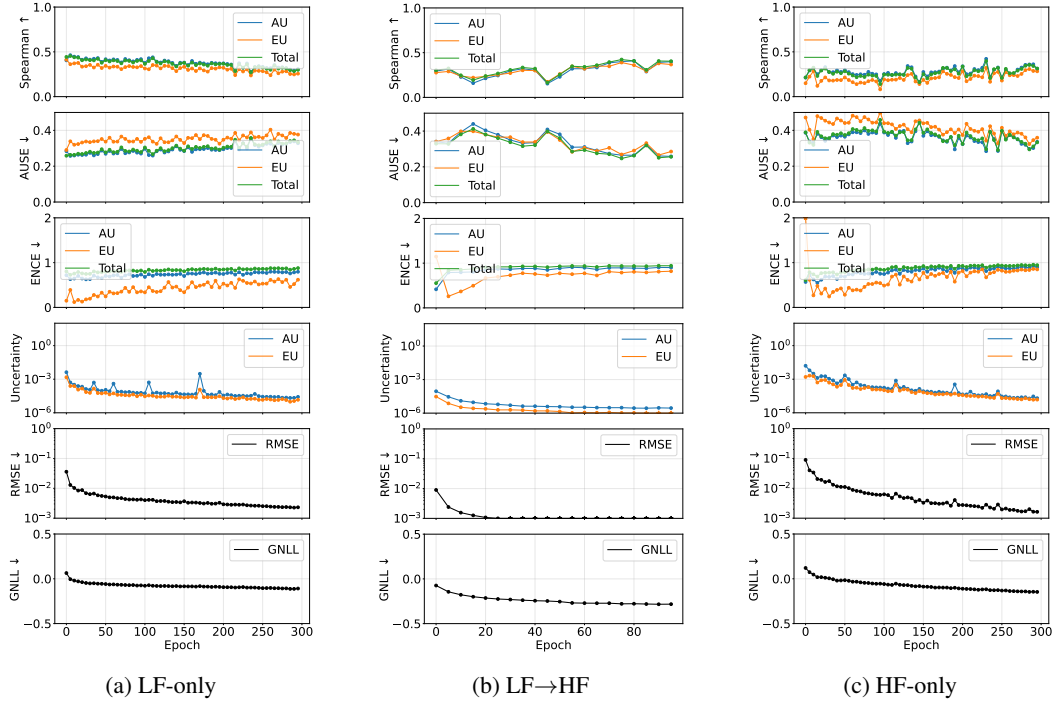


Figure 13: Training dynamics of uncertainty under energy-based shift, evaluated on the training set (Train-ID). Columns compare LF-only, LF→HF, and HF-only training.

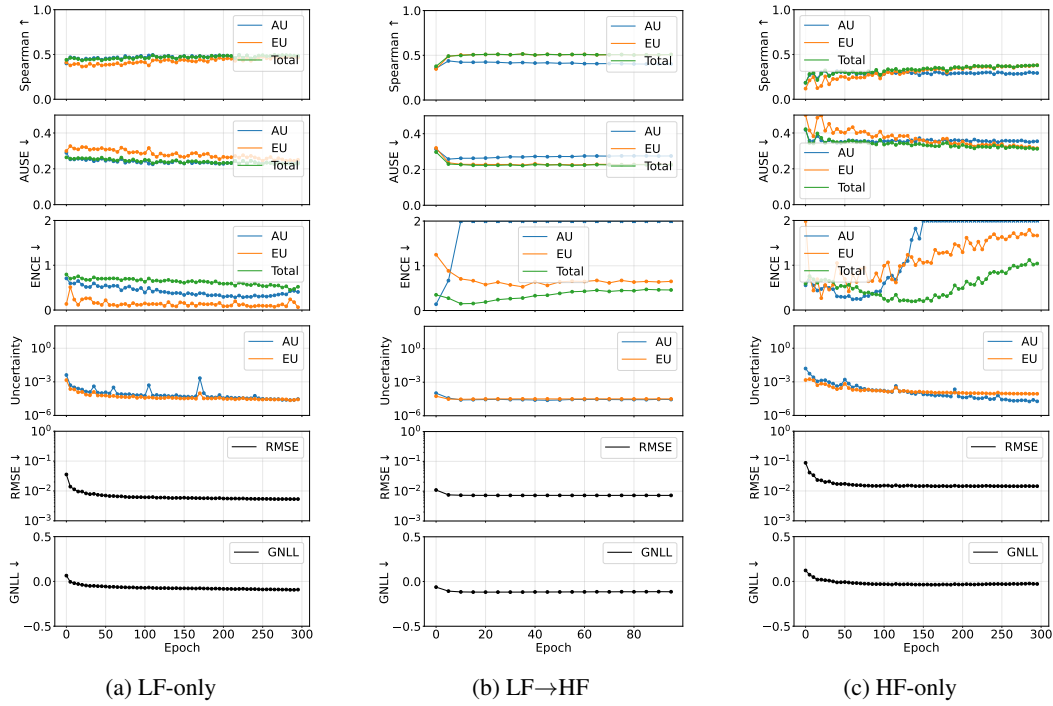


Figure 14: Training dynamics of uncertainty under energy-based shift, evaluated on the in-domain test set (Test-ID). Columns compare LF-only, LF→HF, and HF-only training.

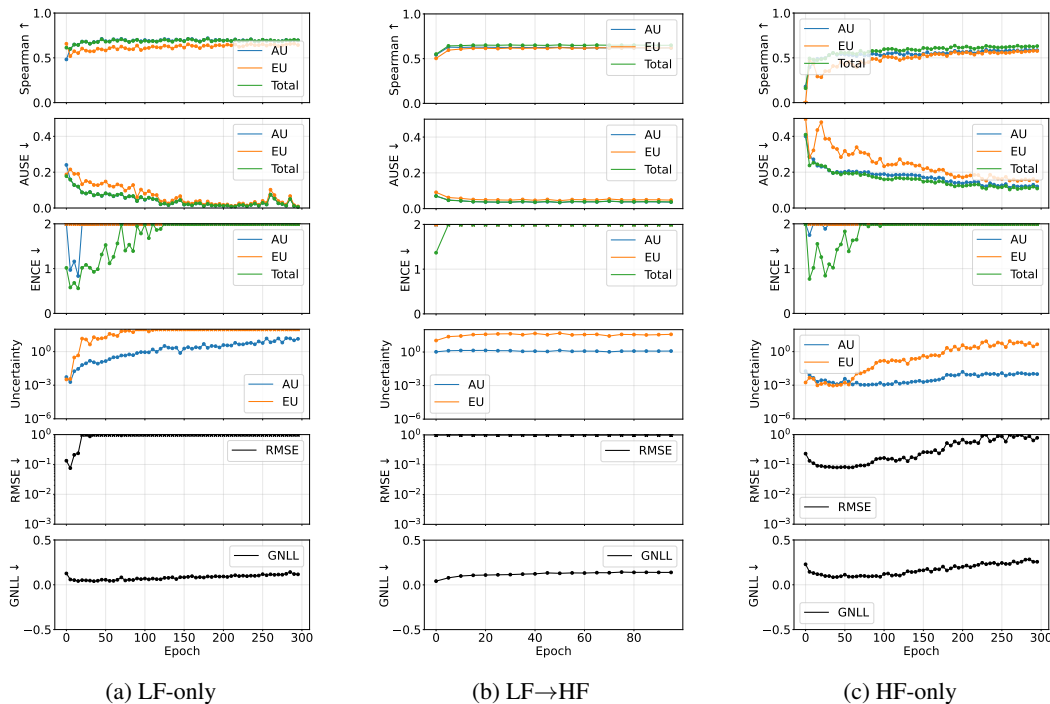


Figure 15: Training dynamics of uncertainty under energy-based shift, evaluated on the out-of-domain test set (Test-OOD). Columns compare LF-only, LF→HF, and HF-only training.

I Calibrated reliability diagrams

Figures 16 and 17 report reliability diagrams after post-hoc calibration using isotonic regression fitted on the validation set. As expected, calibration substantially improves reliability on the in-domain test set (Test-ID), where the predicted uncertainty aligns closely with the observed error across all training regimes. In this setting, calibration effectively corrects the scale of uncertainty without altering its ranking properties.

However, the effect of calibration differs markedly across distributional shifts. Under system-level shift (Figure 16), the calibrated reliability diagrams remain qualitatively similar to the uncalibrated ones, indicating that the calibration mapping learned on in-domain data transfers reasonably well to unseen molecular systems. In contrast, under energy-based shift (Figure 17), calibration fails to generalize: the mapping learned on low-energy configurations leads to severe miscalibration on high-energy configurations, where predicted uncertainties no longer match the scale of the observed errors.

This highlights a key limitation of post-hoc calibration: while it can improve in-domain reliability, it assumes that the relationship between uncertainty and error is stationary across regimes. In energy-based OOD settings, where error distributions and uncertainty magnitudes shift significantly, this assumption breaks down. As a result, calibrated uncertainty can become less reliable than the uncalibrated signal, despite preserving its ranking properties. These results reinforce the need to evaluate uncertainty not only in terms of calibration on validation data, but also in terms of robustness under distributional shift.

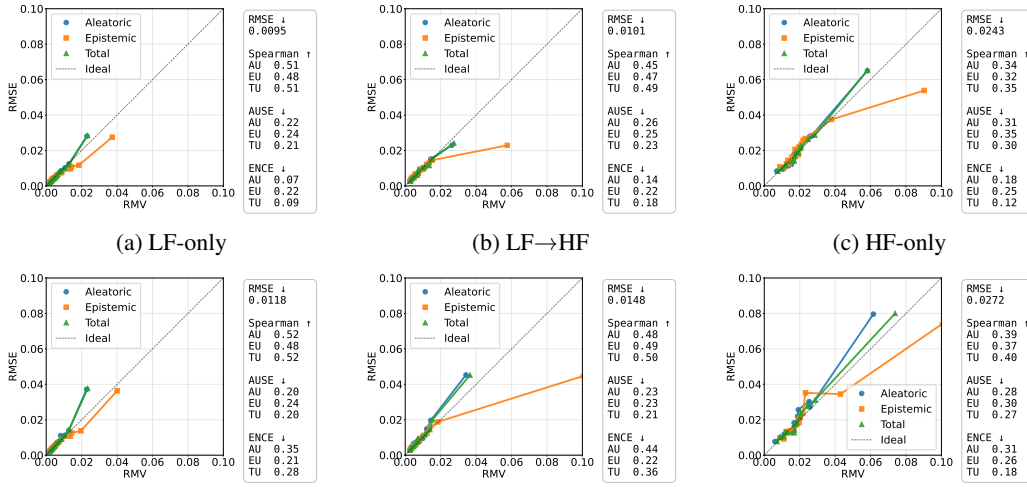


Figure 16: Calibrated reliability diagrams under system-level shift. Top row: in-domain test set (Test-ID). Bottom row: out-of-domain test set (Test-OOD). Columns correspond to LF-only, LF→HF, and HF-only training.

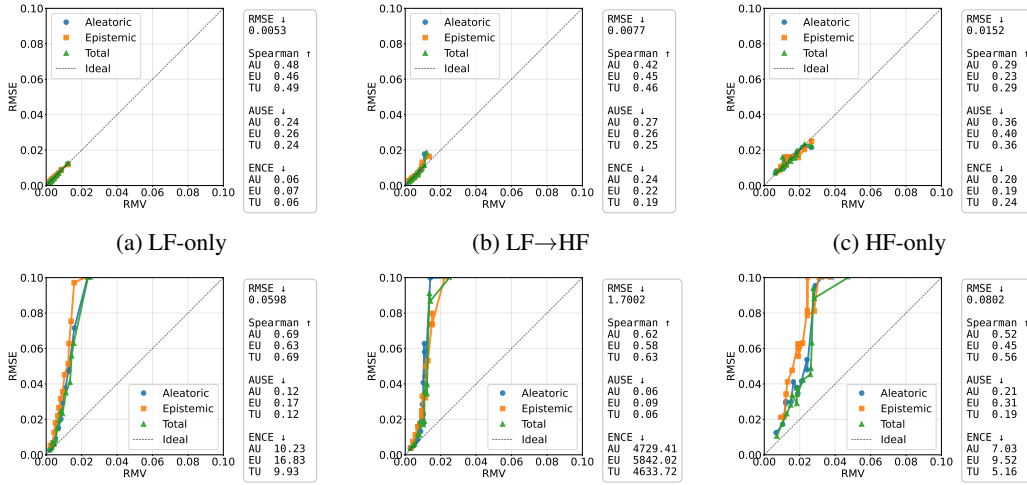


Figure 17: Calibrated reliability diagrams under energy-based shift. Top row: in-domain test set (Test-ID). Bottom row: out-of-domain test set (Test-OOD). Columns correspond to LF-only, LF→HF, and HF-only training.

681 J Computational resources

682 All experiments were conducted on a high-performance computing node equipped with two AMD
683 EPYC 7742 CPUs (128 cores total, 256 threads) with 1TB of total RAM, and 8 NVIDIA A100 GPUs
684 (40GB memory each).

685 Model training and evaluation were performed using GPU acceleration, with an overall computational
686 budget of approximately 500 GPU hours. A single low-fidelity pretraining run requires approximately
687 20 hours on one A100 GPU and occupies about 20GB of GPU memory, while high-fidelity fine-
688 tuning requires approximately 1 hour. The use of deep ensembles, involving multiple independently
689 initialized models, constitutes the primary computational cost of the study.

690 Data preprocessing and auxiliary computations were performed on CPU resources.

691 The full research project required more computing than the experiments reported in the paper due to
692 preliminary or failed experiments.

693 K Limitations and future works

694 We focus only on energy prediction. While energies are central to atomistic modeling and provide
695 a clean setting for uncertainty analysis, practical molecular simulations also require accurate and
696 reliable forces. Extending the analysis to force uncertainty is an important direction for future work,
697 especially because force errors are local, vector-valued, and directly affect molecular dynamics
698 stability.

699 We use mean-variance deep ensembles as our uncertainty estimator. This choice provides a strong,
700 theoretically founded, and widely used baseline for decomposing predictive uncertainty into aleatoric
701 and epistemic components, but it is computationally expensive because it requires training and
702 evaluating multiple models. Future work should investigate whether cheaper alternatives, such as
703 shallow ensembles, last-layer ensembles, multi-head models, evidential methods, or approximate
704 Bayesian approaches, reproduce the same uncertainty dynamics observed here.

705 Our experiments are based on ANI-1x and on controlled system-level and energy-based splits. This
706 design is useful for isolating the effect of data fidelity and distributional shift, but it does not capture
707 the full complexity of realistic simulation workflows, where chemical composition, molecular size,
708 temperature, conformation, and energy can shift simultaneously. Evaluating the same questions
709 on larger and more diverse datasets, including materials and biomolecular systems, would test the
710 generality of our conclusions.

711 L Broader impact

712 This work is primarily methodological and focuses on understanding predictive uncertainty in
713 atomistic machine learning models. As such, it does not directly introduce new capabilities for
714 generating data or making decisions affecting individuals or groups, and its societal impact is indirect.

715 On the positive side, improving the understanding of uncertainty in atomistic models can contribute
716 to more reliable simulations in chemistry and materials science. This has potential downstream
717 benefits in areas such as drug discovery, energy storage, and materials design, where more accurate
718 and trustworthy models can accelerate scientific progress while reducing computational cost. In
719 particular, better uncertainty estimation can enable more efficient data acquisition strategies, reducing
720 the need for expensive quantum-mechanical calculations and lowering the environmental footprint of
721 large-scale simulations.

722 Potential negative impacts are limited but not entirely absent. More efficient and reliable atomistic
723 modeling tools could be used to accelerate the design of harmful chemical compounds or materials
724 if applied inappropriately. Additionally, incorrect or misinterpreted uncertainty estimates could
725 lead to overconfidence in model predictions, potentially resulting in flawed scientific conclusions or
726 unsafe downstream applications. However, our work explicitly highlights the limitations and regime-
727 dependent behavior of uncertainty estimates, which may help mitigate such risks by discouraging
728 overreliance on poorly calibrated models.

729 Overall, we believe the primary impact of this work is to improve the transparency and reliability of
730 uncertainty estimation in atomistic machine learning, contributing to safer and more informed use of
731 these models in scientific and engineering applications.

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